

Topic 3 Chapter 4 Worksheet

In the worksheets for this class, you are going to create MySQL statement for each task described below.

You are required to take a screen shot of the SQL statements and the output for each task.

Paste the screenshot of your results under the numbered question.

Save your work and upload the document to the assignment folder by the due date in the Syllabus.

1. Load and run the script to create the tables. The script is saved in a Microsoft Word document. Copy and paste the code into MySQL Workbench.
2. Similarly, load and run the script to populate the tables.

3. List all the countries and their regions. How many rows returned?

25 rows returned

```
1  -- 3. List all the countries and their regions
2  •  SELECT c.country_name, r.region_name
3      FROM countries c
4      JOIN regions r USING (region_id);
5
```

Result Grid			Filter Rows:	Export:	Wrap C
	country_name	region_name			
►	Belgium	Europe			
	Switzerland	Europe			
	Germany	Europe			
	Denmark	Europe			
	France	Europe			
	Italy	Europe			
	Netherlands	Europe			
	United Kingdom	Europe			
	Argentina	Americas			
	Brazil	Americas			
	Canada	Americas			
	Mexico	Americas			
	United States ...	Americas			
	Australia	Asia			
	China	Asia			
	India	Asia			
	Japan	Asia			
	Malaysia	Asia			
	Singapore	Asia			
	Egypt	Middle East and Africa			
	Israel	Middle East and Africa			
	Kuwait	Middle East and Africa			
	Nigeria	Middle East and Africa			
	Zambia	Middle East and Africa			
	Zimbabwe	Middle East and Africa			

4. What products are in inventory, how much inventory is there, and what are their categories?

For a product to be in inventory, the quantity must be greater than zero. 3 tables will be used. The default 1000 rows will be returned (unless you have changed the default).

```
9 • SELECT p.product_name, pc.category_name, i.quantity
10 FROM products p
11 JOIN inventories i USING (product_id)
12 JOIN product_categories pc USING (category_id)
13 WHERE i.quantity > 0;
```

Result Grid			
Filter Rows: Export: Wrap Cell Contents			
	product_name	category_name	quantity
▶	Intel Xeon E5-2697 V4	CPU	100
	Intel Xeon E5-2697 V4	CPU	71
	Intel Xeon E5-2697 V4	CPU	58
	Intel Xeon E5-2697 V4	CPU	46
	Intel Xeon E5-2697 V4	CPU	34
	Intel Xeon E5-2697 V4	CPU	23
	Intel Xeon E5-1650 V4	CPU	97
	Intel Xeon E5-1650 V4	CPU	67
	Intel Xeon E5-1650 V4	CPU	30
	Intel Xeon E5-1650 V4	CPU	50
	Intel Xeon E5-1650 V4	CPU	36
	Intel Xeon E5-2640 V4	CPU	97
	Intel Xeon E5-2640 V4	CPU	67
	Intel Xeon E5-2640 V4	CPU	31
	Intel Xeon E5-2640 V4	CPU	50
	Intel Xeon E5-2640 V4	CPU	36
	Intel Core i7-6950X (...	CPU	101
	Intel Core i7-6950X (...	CPU	71
	Intel Core i7-6950X (...	CPU	35
	Intel Core i7-6950X (...	CPU	54
	Intel Core i7-6950X (...	CPU	2

5. What products are in inventory, where are they located in inventory, and what are their categories?

Exclude products where the state is NULL. Order by state. 925 rows returned.

```

17 • SELECT p.product_name, pc.category_name, l.state
18 FROM products p
19 JOIN inventories i USING (product_id)
20 JOIN product_categories pc USING (category_id)
21 JOIN warehouses w USING (warehouse_id)
22 JOIN locations l USING (location_id)
23 WHERE i.quantity > 0
24 AND l.state IS NOT NULL
25 ORDER BY l.state;

```

Result Grid   Filter Rows: Export:  Wrap Cell Content			
	product_name	category_name	state
▶	PNY VCGGTX1080T11XGPB-OC	Video Card	California
	Supermicro X10SDV-8C-TLN4F	Mother Board	California
	Western Digital WD2003FZEX	Storage	California
	Asus X99-E-10G WS	Mother Board	California
	PNY VCQM6000-PB	Video Card	California
	Intel Xeon E5-1650 V4	CPU	California
	ASRock EP2C602-4L/D16	Mother Board	California
	Intel DG43RK	Mother Board	California
	Corsair CB-9060011-WW	Video Card	California
	Intel Xeon E5-2640 V4	CPU	California
	MSI GeForce GTX 1080 TI AR...	Video Card	California
	Asus X99-E WS/USB 3.1	Mother Board	California
	Corsair Vengeance LPX	Storage	California
	Intel Core i7-6950X (OEM/Tray)	CPU	California
	Samsung MZ-V6E500	Storage	California
	Gigabyte X299 AORUS Gaming 3	Mother Board	California
	Kingston	Storage	California
	Intel Core i7-3960X Extreme ...	CPU	California
	Samsung MZ-75E1T0B/AM	Storage	California
	MSI GTX 1080 TI LIGHTNING X	Video Card	California
	Supermicro X10SAT-O	Mother Board	California
	MSI X299 TOMAHAWK ARCTIC	Mother Board	California
	Kingston SA400S37/120G	Storage	California

6. What products have customers purchased?

665 rows returned.

```

28 • SELECT c.name, p.product_name
29 FROM products p
30 JOIN order_items USING (product_id)
31 JOIN orders USING (order_id)
32 JOIN customers c USING (customer_id);

```

Result Grid		Filter Rows:	Export:
	name	product_name	
▶	Raytheon	G.Skill Trident X	
	Raytheon	G.Skill Ripjaws V Series	
	Raytheon	G.Skill Ripjaws 4 Series	
	Raytheon	Intel Xeon E5-2643 V2 (OEM/Tray)	
	Raytheon	G.Skill Ripjaws V Series	
	Raytheon	Intel Xeon E5-2695 V4	
	Raytheon	Asus X99-E-10G WS	
	Raytheon	Kingston	
	Raytheon	Seagate ST31000340NS - FFP	
	Raytheon	Intel Xeon E5-2660 V4	
	Raytheon	Intel Xeon E5-2697 V4	
	Raytheon	MSI X99A GODLIKE GAMING CAR...	
	Raytheon	Seagate ST2000DX002	
	Raytheon	PNY VCQK5200-PB	
	Raytheon	Intel Xeon E5-2660 V4	
	Raytheon	Intel Xeon E5-1650 V3	
	Raytheon	ATI FirePro R5000	
	Raytheon	AMD 100-505989	
	Raytheon	ASRock EP2C612 WS	
	Raytheon	PNY SSD7CS1311-120-RB	
	Raytheon	Gigabyte GV-N108TAORUSX W-1...	
	Raytheon	Supermicro MBD-X10DAX	
	Raytheon	Intel Xeon E5-1680 V3 (OEM/Tray)	
	Raytheon	MSI X99A GODLIKE GAMING	
	Raytheon	G.Skill Ripjaws V Series	
	Raytheon	EVGA 11G-P4-6598-KR	
	Raytheon	Asus GTX780TI-3GD5	
	Plains GP...	MSI GeForce GTX 1080 Ti GAMIN...	

Now show just the names of the products that have been purchased.
Order by product name. 216 rows returned.

```

36 • SELECT distinct p.product_name
37 FROM products p
38 JOIN order_items USING (product_id)
39 ORDER BY p.product_name;

```

Result Grid		Filter Rows:	Export:
	product_name		
▶	ADATA ASU800SS-128GT-C		
	ADATA ASU800SS-512GT-C		
	AMD 100-505989		
	AMD 100-506061		
	AMD FirePro S7000		
	AMD FirePro W9100		
	AMD Opteron 6378		
	ASRock C2750D4I		
	ASRock EP2C602-4L/D16		
	ASRock EP2C612 WS		
	ASRock Fatal1ty X299 Prof...		
	ASRock X299 Taichi		
	ASRock X99 Extreme11		
	Asus GTX780TI-3GD5		
	Asus KGPE-D16		
	Asus MAXIMUS IX CODE		
	Asus MAXIMUS IX FORMULA		
	Asus MAXIMUS VIII EXTRE...		
	Asus PRIME X299-A		
	Asus PRIME X299-DELUXE		
	Asus Rampage V Edition 10		
	Asus RAMPAGE V EXTREME		
	Asus ROG MAXIMUS IX EX...		
	Asus ROG STRIX X99 GAMI...		
	Asus ROG-POSEIDON-GTX...		
	Asus Sabertooth 990FX		
	Asus SABERTOOTH X99		
	Asus STRIX X299-E GAMING		

7. Of the items sold, which warehouses supplied the items?
Order by warehouse name.

```

43 • SELECT distinct w.warehouse_name
44 FROM orders o
45 JOIN order_items USING (order_id)
46 JOIN inventories i USING (product_id)
47 JOIN warehouses w USING (warehouse_id)
48 ORDER BY w.warehouse_name;

```

Result Grid		Filter Rows:	Export:	Wrap C
	warehouse_name			
▶	Beijing			
	Bombay			
	Mexico City			
	New Jersey			
	San Francisco			
	Seattle, Washington			
	Southlake, Texas			
	Sydney			
	Toronto			

8. Which employees have sold something?

9 rows returned.

```

51 • SELECT distinct e.first_name, e.last_name
52 FROM employees e
53 JOIN orders o ON e.employee_id = o.salesman_id
54 JOIN order_items oi ON o.order_id = oi.order_id;

```

Result Grid		Filter Rows:	Export:	Wrap Cell Content:
	first_name	last_name		
▶	Lily	Fisher		
	Grace	Ellis		
	Evie	Harrison		
	Scarlett	Gibson		
	Chloe	Cruz		
	Isabelle	Marshall		
	Daisy	Ortiz		
	Freya	Gomez		
	Florence	Freeman		

9. Which employees do not have orders?

Order the list by last name. 97 rows returned.

```

58 • SELECT e.first_name, e.last_name, o.order_id, o.salesman_id, e.employee_id
59 FROM employees e
60     LEFT JOIN orders o ON e.employee_id = o.salesman_id
61 WHERE o.salesman_id is NULL
62 ORDER BY e.last_name;

```

Result Grid |   Filter Rows: | Export:  | Wrap Cell Content: 

first_name	last_name	order_id	salesman_id	employee_id
Ibrahim	Alexander	NULL	NULL	37
Tommy	Bailey	NULL	NULL	1
Dexter	Barnes	NULL	NULL	20
Ollie	Bennett	NULL	NULL	18
Eliza	Black	NULL	NULL	90
Imogen	Boyd	NULL	NULL	75
Elliot	Brooks	NULL	NULL	14
Felix	Bryant	NULL	NULL	36
Ivy	Burns	NULL	NULL	83
Jamie	Butler	NULL	NULL	32
Isabella	Cole	NULL	NULL	49
Jackson	Coleman	NULL	NULL	23
Blake	Cooper	NULL	NULL	3
Nathan	Cox	NULL	NULL	5
Lucy	Crawford	NULL	NULL	73
Lilly	Daniels	NULL	NULL	91
Caleb	Diaz	NULL	NULL	40
Elizabeth	Dixon	NULL	NULL	80
Annabelle	Dunn	NULL	NULL	101
Harriet	Ferguson	NULL	NULL	97
Austin	Flores	NULL	NULL	30
Olivia	Ford	NULL	NULL	43
Seth	Foster	NULL	NULL	34
Gracie	Gardner	NULL	NULL	105

10. Which products do not have orders?
 32 rows returned.

```

65 • SELECT p.product_name, oi.order_id
66 FROM products p
67 LEFT JOIN order_items oi USING (product_id)
68 WHERE order_id is NULL;

```

Result Grid |  Filter Rows: | Export:  | Wrap C

	product_name	order_id
▶	Seagate ST3000DM008	NULL
	Samsung MZ-75E500B/AM	NULL
	Corsair Dominator Platinum	NULL
	Intel Xeon E5-2695 V3 (OEM/Tray)	NULL
	Western Digital WD20EZRZ	NULL
	Intel Core i7-5960X (OEM/Tray)	NULL
	Intel Xeon E5-2650 V4	NULL
	Gigabyte GV-N98TWF30C-6GD	NULL
	Corsair Vengeance Pro	NULL
	G.Skill Ripjaws V Series	NULL
	PNY VCQP6000-PB	NULL
	MSI X99A WORKSTATION	NULL
	MSI Z270 XPOWER GAMING TITA...	NULL
	AMD 100-5056062	NULL
	Intel Xeon E5-2470V2	NULL
	ASRock Z270 SuperCarrier	NULL
	NVIDIA VCQM4000-PB	NULL
	Asus Z10PE-D8 WS	NULL
	G.Skill Trident Z RGB	NULL
	Intel Core i7-980	NULL
	Intel Xeon E5-2680 V4	NULL
	Zotac ZT-P10810C-10P	NULL
	MSI GeForce GTX 1080 TI ARMOR...	NULL
	Intel Xeon E5-2699 V3 (OEM/Tray)	NULL
	Seagate ST10000DM0004	NULL
	Western Digital WD1003FZEX	NULL
	Intel Xeon E5-2643 V4 (OEM/Tray)	NULL
	Crucial	NULL